SYED MAHFUZUL MAZID

TECH LEAD | SOFTWARE ARCHITECTURE | FULL-STACK & BACKEND ENGINEERING

PHP | LARAVEL | PYTHON | FASTAPI | FLASK | VUE.JS | NEXT.JS | REST APIs | MICROSERVICES | AWS | DOCKER | CI/CD

Guwahati, India | Contact via LinkedIn

LinkedIn: linkedin.com/in/syed-mahfuzul-mazid | **Portfolio**: smahfuzm.in | **GitHub**: github.com/mahfuzt786

Open to Relocation - Bengaluru | Hyderabad | Pune | Kolkata | Mumbai | Kochi | Remote (Pan-India)

PROFESSIONAL SUMMARY

Tech Lead and Software Analyst with 13+ years of experience as both a PHP/Laravel Developer and a Python Developer (FastAPI, Flask), architecting and delivering enterprise-grade REST APIs, microservices, and full-stack web platforms for HealthTech, AI/NLP, and E-commerce clients in Germany, Switzerland, and the USA. Deep hands-on expertise across Laravel, Vue.js, and Next.js on the PHP side and FastAPI/Flask on the Python side, backed by strong command of Docker, CI/CD, Git, secure coding, and DevOps practices. Proven record of leading and mentoring cross-functional engineering teams of 4-15+ developers - owning architecture, coding standards, and code review - and driving legacy-system modernisation that reduced operating costs by 30% and cut manual labour by 40-50%. Strong fit for **enterprise architecture and technical leadership** roles at **India's product, services, and global-capability-center companies**.

CORE TECHNICAL SKILLS

Languages: Python, PHP, JavaScript (ES6+), TypeScript, HTML5, CSS3, SQL

Frameworks & Libraries: Laravel, FastAPI, Flask, Vue.js, Next.js, React, jQuery, Electron, Cordova, MVC, OOP

Databases: MySQL, PostgreSQL, Firebase

APIs & Backend: RESTful APIs, SOAP APIs, Microservices, Nginx, Stripe, Pusher / Laravel Echo, Server-Sent Events (SSE), Laravel Sanctum

DevOps & Cloud: Docker, CI/CD, AWS, Git, SVN, Linux, Deployment Automation

AI / ML: Natural Language Processing (NLP), Chatbot Development, Image Recognition, ML Pipelines

Testing: PHPUnit, Pytest, Unit Testing, Integration Testing, API Testing, Automated Testing

AWS: EC2, S3, RDS

Architecture: Software Architecture, System Design, Solution Architecture, API Design, Database Design, Scalability, Performance Optimization, Design Patterns, Legacy Modernization, Secure Coding

Leadership: Team Leadership, Agile / Scrum, Code Review, Mentoring, Cross-Functional Collaboration, Stakeholder Management, Technical Leadership

Languages Spoken: English (Professional), Hindi (Full Professional), Assamese (Native), Bengali (Limited)

PROFESSIONAL EXPERIENCE

Alegra Labs GmbH - Guwahati, India | Aug 2013 - Present

Single-employer career growth: Junior Developer -> Senior Developer -> Lead Developer -> Tech Lead & Software Analyst. Delivered 7 production projects for international clients across Germany, Switzerland, and the USA.

Tech Lead & Software Analyst - Alegra Labs GmbH **Dec 2020 – Present**

Technical Leadership | Software Architecture | Full-Stack & Backend Engineering

Project: Dentapal - Healthcare & Clinic Operations SaaS | Dentapal GmbH, Berlin, Germany

- **Architected** a multi-portal SaaS platform (Admin, Employee, Customer portals) for dental clinic operations using Next.js, React, and TypeScript on the frontend and Laravel+ MySQL on the backend, with Sanctum authentication and role-based access control.
- **Automated** the HKP (Heil- und Kostenplan) dental treatment plan workflow with a hybrid rules-and-ML system, reducing clinician labour by 40-50% and improving patient throughput.
- **Led** a cross-functional team of 4 developers; directed architecture, enforced coding standards, mentored junior engineers, and orchestrated migration of legacy Python systems to Laravel / MySQL.
- **Built** an appointment dispatch engine with calendar and list views, conflict-aware scheduling, bulk actions, ISO week / month filtering, and a vacation / sick-day request workflow with multi-stage approvals.
- **Integrated** Stripe billing, Pipedrive CRM, BuchhaltungsButler accounting, and Pusher / Laravel Echo for real-time messaging; built a Windows desktop connector (DS-Win) for revenue and turnover-statistics synchronisation.
- **Delivered** DentaBot, a FastAPI-based AI service for clinical chat and proposal generation; shipped CI/CD pipelines, Docker-based deployments, and operational KPI dashboards with scoped APIs and Server-Sent Events.

Lead Developer - NLP Transcription Platform - Alegra Labs GmbH **May 2020 - Dec 2020**

Project: Interscriber | Zurich University of Applied Sciences (ZHAW), Switzerland

- **Built** an audio-to-text transcription platform leveraging advanced NLP techniques in collaboration with ZHAW researchers, automating interview transcription end-to-end.
- **Engineered** the system using Electron with Vue.js on the frontend and Python, Flask, Nginx, and PostgreSQL on the backend, exposed via REST APIs.
- **Reduced** operating costs by 30% through legacy-codebase modernisation and cut manual labour by 40% through pipeline automation.
- **Coordinated** a multi-national engineering team, aligning database timelines and delivery milestones across mixed seniority levels.

Lead Developer - Conversational AI / Chatbot - Alegra Labs GmbH **Dec 2019 - May 2020**

Project: Machine Learning Chatbot Platform

- **Designed** and implemented conversational agents and backend services using Python, Flask, and MySQL, focused on intent recognition and entity extraction.
- **Delivered** scalable REST APIs in Flask enabling drop-in JavaScript integration with client websites.
- **Trained** QA teams to build and refine intent models, improving response accuracy and end-user interaction quality.
- **Authored** reusable, distributed-cloud-ready code and contributed to proposal design for prospective clients.

Lead Developer - Image Recognition Web App - Alegra Labs GmbH **Apr 2019 - Dec 2019**

Project: Poster Recognition Platform - nikolaus-spitzer.de, Germany

- **Developed** an ML-driven image-recognition system in Python, Flask, and jQuery to detect and verify street posters from mobile photos under varying lighting and angles.
- **Architected** data-engineering pipelines that improved ML throughput and recognition accuracy.
- **Eliminated** thousands of hours of manual verification work for the client through automation, materially improving project economics.

- **Created** a product-health monitoring framework combining end-user satisfaction signals with technical fault statistics for ongoing reliability.

Senior Developer - CMS & API Integrations - Alegra Labs GmbH **Apr 2018 - Apr 2019**

Project: DigiPro Media CMS - Florida, USA (vacation-rental properties)

- **Built** a multi-module CMS using PHP 7, HTML5, jQuery, and the brisk-coder PHP MVC framework, deployed across homeownerscollection.com, premierisland.com, and portofinoisland.com.
- **Integrated** multiple SOAP and REST APIs (Escapia, IQware, Propay) covering room booking, payments, and channel synchronisation.
- **Collaborated** remotely with an international team of 15+, owning coding tasks across CMS modules and leading large-scale deployments.

Senior Developer - E-commerce & ERP Platform - Alegra Labs GmbH **Jan 2014 - Mar 2018**

Project: Kamwo Herbs - New York City, USA (Chinese Herbal Medicine)

- **Engineered** a large-scale E-commerce and ERP platform for manufacturing and shipping (PHP 7, HTML5, MySQL, Firebase), integrating warehousing, shipping, and logistics modules.
- **Delivered** a hybrid mobile shopping app (jQuery Mobile, Cordova) on Google Play and launched the practitioner-search portal acudirect.com and the online herb guide at kamwo.com.
- **Improved** application performance by 30% through code optimisation and increased client satisfaction by 20% through proactive project management.

Junior Developer - SVG Manipulation Module - Alegra Labs GmbH **Aug 2013 - Dec 2013**

Project: Picletta - Facebook Application

- **Developed** an SVG manipulation module supporting resize, rotate, merge, animation, embedded YouTube playback, and shape conversion using JavaScript and jQuery.
- **Contributed** across the full SDLC - requirements gathering, design, implementation, and production release; translated customer requirements into actionable use cases.

EDUCATION

MCA, Computer Science - St. Anthony's College, Shillong | CGPA 7.46 / 10 2010 - 2013

BCA, Computer Science - Dibrugarh University, Dibrugarh | 64% 2007 - 2010

CERTIFICATIONS & AWARDS

Awards: Anandaram Baruah Award (contribution to technology and education) | Jigyas Talent Search Examination (academic excellence)

PUBLICATIONS

The Architect's Journey - Six-volume programming book series covering programming fundamentals, desktop and database applications with Python, Java, and SQL, web development, AI/ML, system design, and advanced computation; published on Amazon Kindle (free to read at https://smahfuzm.in/book).